

Title

Augmented Reality Enhanced by Mirror Visual Feedback for Upper Limb Recovery Post-Stroke:
A Multigroup Randomised Rehabilitation Trial

Abstract

Forty-four stroke survivors with moderate-to-severe arm and hand weakness were randomised to receive one of three rehabilitation approaches over six weeks: mirror visual feedback combined with interactive augmented reality exercises, augmented reality alone, or conventional physical therapy. Each intervention included 18 sessions of 90-minute duration performed three times weekly. Objective motor recovery was measured using standardised clinical scales assessing upper limb movement quality, functional strength, and postural stability. Sensory testing evaluated proprioception and tactile perception. Additionally, participants completed self-report questionnaires regarding functional arm use in daily activities and quality-of-life perceptions. The combined mirror-plus-augmented-reality group demonstrated significantly greater improvements in motor control and balance capacity compared to conventional treatment. The augmented-reality-only group also exceeded conventional therapy on these measures. Within interactive technology groups, the mirror-enhanced approach showed superior outcomes for fine motor dexterity and touch sensation. No therapy-related injuries or adverse events occurred. Improvements were sustained at three-month follow-up assessment in all groups. These results suggest that combining mirror visual principles with interactive digital environments produces synergistic rehabilitation benefit for stroke survivors with moderate motor impairment.